

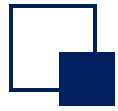


Chapter 11: Generic and Reflection



Outline

- Why generics? What is generic?
- Generic classes, generic methods, and generic interfaces
- Generic wildcards
- Generic design - template method pattern
- Reflection
- Design security of singleton pattern



Why do you need generics?

❑ Consider the following code for problems

```
ArrayList list = new ArrayList();  
list.add("str");  
list.add(100);  
list.add(true);  
String name = (String) list.get(0);
```

1. No error checking, you can add any object to the array list.
2. Forced type conversion must be done to get a value.

What is generic?

❑ Java generics are a new feature introduced in Java 5 that provides compile-time type safety monitoring that allows us to detect illegal data structures **at compile time**.

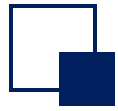
- Introduce type parameters, which specifies the data type of all operations as a parameter.

For example, specify a generic String

```
ArrayList<String> list = new ArrayList<>();  
list.add("str");  
String name = list.get(0);
```

```
ArrayList list = new ArrayList<>();  
list.add("str");  
String name = (String) list.get(0);
```

When the get method is called, there is no need for forced type conversion, and the compiler knows that the return value is of type String, not Object.



What is generic?

❑ Consider whether the following is feasible

```
ArrayList<String> list = new ArrayList<>();  
list.add(100);
```

- For no reason, the compiler prompts

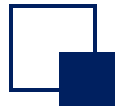
```
Required type: String  
Provided: int
```

The compiler knows<String> that the add method in List has a parameter of type String, which is safer than using the Object type. Illegal data types detected during the compilation phase.



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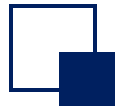
Generic Classes (1)

❑ A generic class is a class with one or more type variables.

- Syntax format: Declare a generic after the class name, and the generic class declares the type variable before using it. Type variables usually use shorter capitalizations such as T, E, K, V, etc.

Declare a single generic

```
class Generic<T> {                                //Generic type of declarative class <T>
    public T t;                                    //The type of variable t is generic T
    public Generic(){}
    public Generic(T t){                          //The type of incoming parameter is T
        this.t = t;
    }
    public void fun1(T t) {}
    public T fun2(T t) {}                        //The function returns a value of type T
}
```



Generic Classes (2)

□ Type parameters (also known as type variables) are placed as placeholders to indicate the type of assignment

- E: (Element) element
- K: (Key) key
- N: (Number) Number
- T: (Type) type
- V: (Value) value
- S, U, V, etc.: Types 2, 3, 4 in multiparameter situations

```
public class Generic<T>{  
    .....  
}
```




Generic Classes (3)

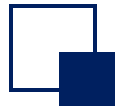
- A single generic class is used

```
class Generic<T> {  
    public T t;  
    public void fun(T t) {}  
}
```

```
Generic<String> g1 = new Generic<>();  
g1.t = "str";           //This is the legal data type  
g1.t = 100;             //This is the illegal data type
```



```
Generic<Integer> g2 = new Generic<>();  
g2.t = 100;             //This is the legal data type
```

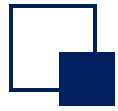


Generic Classes (4)

- Multiple generic classes are used

```
class Generic<T, E> {  
    public T t;  
    public E e;  
    public void fun(T t, E e) {}  
}  
  
Generic<String, Integer> g3 = new Generic<>();  
g3.t = "str";  
g3.e = 100;
```

- Note that generics cannot be used for basic data types, such as int
Generic<int> g = new Generic<>(); ❌
Generic<Integer> g = new Generic<>();



Generic Method (1)

- ❑ Generic methods can be defined in a normal class or in a generic class. A method that uses generic members in a generic class is not a generic method, only a method that declares a generic is a generic method.

Note that the following is not a generic method

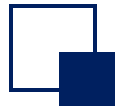
```
class Generic<T> {  
    public T t;  
    public void func(T t) {}  
}
```



Generic methods are not explicitly declared

Note that the following is a generic method

```
class Generic<T> {  
    public T t;  
    public <T> void func(T t) {}  
}
```



Generic Methods (2)

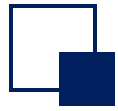
Generic methods are defined in the non-generic class

```
class GenericFun{  
    public <T,E> void fun1(E e){}  
    public <T> T fun2(T t){  
        return t;  
    }  
}
```

- When calling a generic method, fill in the specific type in parentheses in the method name.

```
GenericFun g = new GenericFun();  
g.<String>fun2("str");
```

- In most cases, method calls can omit type parameters.
g.fun2("str");//The return value is str



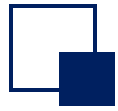
Generic Methods (3)

Generic methods are defined in the generic class

```
class GenericFun<K>{  
    public <T> T fun2(T t, K k){  
        return null;  
    }  
}
```

- With the generic method, the incoming argument here must match the type declared by the generic class

```
GenericFun<Integer> g = new GenericFun<>();  
g.fun2("str1", 123) ;
```

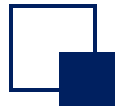


Generic Methods (4)

- ❑ If the generic of a generic method matches the generic name declared by the generic class, the generic in the generic method overrides the generic of the class.

```
class GenericFun<K>{  
    public <T,K> T fun2(T t, K k){  
        return null;  
    }  
}
```

- Using the generic method, the generic of the class is Integer, and the parameters passed by the generic method are two Strings
 GenericFun<Integer> g = new GenericFun<>();
 g.fun2("str1","str2") ;

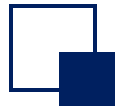


Generic Methods (5)

- ❑ The static generic method of the class must not use the generics declared in the generic class, and can be declared independently.

```
class GenericStaticMethod<K>{  
    private K k;  
    public GenericStaticMethod(K k){  
        this.k=k;  
    }  
    public static GenericStaticMethod <K> fun1(K k){  
        return new GenericStaticMethod <K>(k)}  
} //Error: Unable to reference non-static variable K from static context  
public static <K> GenericStaticMethod<K> fun1(K k){  
    return new GenericStaticMethod<K>(k)}  
} //Compilation is successful  
public static <T> GenericStaticMethod<T> fun1(T t){  
    return new GenericStaticMethod<T>(t)}  
} //Written in another type
```





Generic method

// Generic declarations are used when defining the class Durian

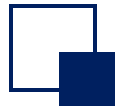
```
class Durian<T> {
```

```
    // Use the T-type parameter to define instance properties  
    private T info;
```

```
    // The following method uses the T-type parameter to define the constructor  
    public Durian(T info) { this.info = info; }  
    public void setInfo(T info) { this.info = info;}  
    public T getInfo() {    return this.info;}
```

```
    // Static generic methods should be distinguished using other types:  
    public static <K> Durian<K>readyear(K info) {  
        return new Durian<K>(info);  
    }
```

```
}
```

Generic method

```
public class GenericDurian {  
    public static void main(String[] args) {
```

// Since the T-parameter is passed to String, the constructor parameter can only be String

```
Durian<String> a1 = new Durian<>("Musang King");//name
```

```
System.out.println(a1.getInfo());
```

// Since the T-parameter is passed a Double, the constructor parameter can only be a Double

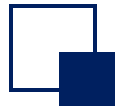
```
Durian<Double> a2 = new Durian<>(1.23);//weight
```

```
System.out.println(a2.getInfo());
```

```
Durian<Integer> a3 = Durian.readyear(2025);
```

```
System.out.println(a3.getInfo());
```

```
    }  
    public static <K> Durian<K>readyear(K info) {  
        return new Durian<K>(info);  
    }  
}
```



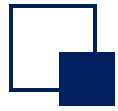
Generic Interfaces (1)

Define the following generic interfaces

```
interface GenericInterface <T>{  
    T fun1();  
}
```

The implementation class is a non-generic class, and the generic type of the interface must be specified

```
class GenericInterfaceImpl implements GenericInterface<String> {  
    @Override  
    public String fun1() {  
        return null;  
    }  
}
```



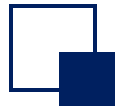
Generic Interfaces(2)

Define the following generic interfaces

```
interface GenericInterface <T>{  
    T fun1();  
}
```

The implementation class is a generic class, and the generic of the implementation class should be consistent with the interface

```
class GenericInterfaceImpl<T> implements GenericInterface<T> {  
    @Override  
    public T fun1() {  
        return null;  
    }  
}
```



Generic interfaces

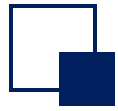
```
interface ShowInterface<T> { public void show(T t); }
```

//The implementation class is non-generic

```
class ShowClass1 implements ShowInterface<String>{  
    public void show(String t){  
        System.out.println("show:"+t);  
    }  
}
```

//The implementation class is generic

```
Class ShowClass2<T> implements ShowInterface<T>{  
    public void show(T t){  
        System.out.println("show:"+t);  
    }  
}
```



Generic interfaces

```
public static void main(String[] args) {  
    //The implementation class is non-generic  
    ShowClass1 obj = new ShowClass1();  
    obj.show("java");  
  
    //The implementation class is generic, type is determined when used  
    ShowClass2<Integer> obj = new ShowClass2<>();  
    obj.show(6);  
}
```



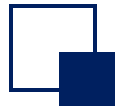
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Generic wildcards

- ❑ A strict generic type system cannot be changed once specified
- ❑ You need to allow type parameters to change
- ❑ At the same time, you need to control the types that can be specified, rather than being unlimited



Generic wildcards

□ Why do you need wildcards

- Consider abstracting a method for all Lists, which `<Integer><String>` can receive and print elements in Lists, regardless of the parameters

```
public static void printAllObject(List<Object> list) {  
    for (Object object : list) {  
        System.out.println(object);  
    }  
}  
  
public static void main(String[] args) {  
    ArrayList<String> list1 = new ArrayList<>();  
    list1.add("java");  
}
```

Object is the parent of all types, but `List<Object>` is not the parent of `List<String>`

Error: `printAllObject(List<Object>)` is not applicable for the arguments(`List<String>`)



Generic wildcards

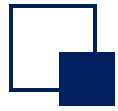
```
public static void printAllObject(List<?> list) {  
    for (Object object : list) {  
        System.out.println(object);  
    }  
}  
  
public static void main(String[] args) {  
    List<String> list1 = new ArrayList<>();  
    list1.add("java");  
    printAllObject(list1);  
    List<Integer> list2 = new ArrayList<>();  
    list2.add(7);  
    printAllObject(list2);  
}
```



Generic wildcards

▣ wildcard

- Allow type parameters to change
- `<? extends ClassName>`: The type argument is a subclass of the `ClassName`
- `<? super ClassName>`: The type parameter is a superclass of `ClassName`
- `<?>`: No limit wildcard

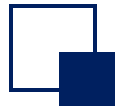


Upper Bound Wildcard

□ <? extends T>

- The class must be a T class, or a subclass of T class

```
public class WildCardExtendsDemo {  
    public static void printAllObject(ArrayList<? extends Number> list) {  
        for (Object object : list) {  
            System.out.println(object);  
        }  
    }  
    public static void main(String[] args) {  
        ArrayList<Double> list1 = new ArrayList<>();  
        list1.add(1.23);  
        printAllObject(list1);  
    }  
}
```



Lower Bound Wildcard

□ <? super T>

- The class must be a T class, or a superclass of T class

```
public class WildCardSuperDemo {  
    public static void printAllObject(ArrayList<? super Double> list) {  
        for (Object object : list) {  
            System.out.println(object);  
        }  
    }  
    public static void main(String[] args) {  
        ArrayList<Number> list1 = new ArrayList<>();  
        list1.add(7);  
        printAllObject(list1);  
    }  
}
```

Type parameter T vs. wildcard?

❑ T represents a definite type

- It is commonly used for the definition of generic classes and generic methods

❑ Wildcard indicates an indeterminate type, not a type variable

- Commonly used for calling codes and parameters for generic methods, not for defining classes and generic methods.

```
T t = operate();  
? a = operate();    //illegal
```



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Template method patterns

□ Template Method

- Define the algorithm skeleton in an operation
- Defer some steps into the subclass
- The template method pattern allows subclasses to redefine certain steps of an algorithm without changing its structure



Template method patterns

❑ AbstractClass

- Abstract templates, define and implement a template method
- Give the skeleton of the top-level logic

❑ ConcreteClass

- Implement one or more abstract methods defined by the parent class

❑ An abstract class can have any number of concrete subclasses

❑ Each abstract class can give different implementations of abstract methods


```
public abstract class AbstractClass
{
    public abstract void PrimitiveOperation1();
    public abstract void PrimitiveOperation2();

    public void TemplateMethod()
    {
        PrimitiveOperation1();
        PrimitiveOperation2();
        System.out.println("");
    }
}
```

Some abstract behaviors are placed in subclass implementations

The template method gives the skeleton of the logic, which is made up of some corresponding abstract operations, all of which are deferred to the subclass implementation

```
public class ConcreteClassA extends AbstractClass
{
    public void PrimitiveOperation1()
    {
        System.out.println("Specific class A method 1 implementation");
    }
    public void PrimitiveOperation2()
    {
        System.out.println("Specific class A method 2 implementation");
    }
}

public class ConcreteClassB extends AbstractClass
{
    public void PrimitiveOperation1()
    {
        System.out.println("Specific class B method 1 implementation");
    }
    public void PrimitiveOperation2()
    {
        System.out.println("Specific class B method 2 implementation");
    }
}
```



Template method patterns

- ❑ Move the common behavior to the superclass to remove the duplicate code in the subclass
- ❑ Improved code reuse
- ❑ Applications
 - Implement the common part of an algorithm at once, leaving the different part to the subclass to implement
 - The behavior common in each subclass should be extracted and centralized into a common parent class to avoid code duplication
 - Control subclass expansion: The template method only calls the "hook" operation at specific points
 - (Hook operation: Provides default behavior, subclasses can be extended if necessary)



Template method patterns

□ participator

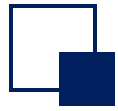
- Abstract and concrete classes

□ collaboration

- Concrete classes rely on abstract classes to implement common steps in algorithms
- Abstract classes rely on concrete classes to implement the specific details of the algorithm

□ effect

- Template methods are especially important in libraries, where they extract public behavior from the library



Template method patterns

❑ The template approach results in a reverse control structure

- Refers to the action of the parent class calling a subclass, not the other way around
- Sometimes referred to as the "Hollywood Rule," i.e. "Don't look for us." We are looking for you"

❑ note

- The template method must specify which operations are hook operations (which can be redefined) and which are abstract operations (which must be redefined)
- Minimize the number of abstract operations that must be redefined when a subclass implements the algorithm



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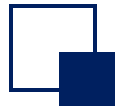


Why d

If there are hundreds or thousands of different X implementations that need to be created
Need to write thousands of if statements to return different X objects?

```
public class Test {  
    interface X {  
        public void test();  
    }  
    class A implements X {  
        @Override  
        public void test() {  
            System.out.println("I am A");  
        }  
    }  
    class B implements X {  
        @Override  
        public void test() {  
            System.out.println("I am B");  
        }  
    }  
}
```

```
public static void main(String[] args) {  
    X a = create1("A");  
    a.test();  
    X b = create1("B");  
    b.test();  
}  
  
public static X create1(String name){  
    if (name.equals("A")) {  
        return new A();  
    }  
    else if(name.equals("B")) {  
        return new B();  
    }  
    return null;  
}
```



Reflection

```
// In the above example, create1 can be rewritten as
public static X create2(String name) {
    Class<?> class = Class.forName(name);
    X x = (X) class.newInstance();
    return x;
}
```

Pass the package name and class name to the create2() method, dynamically load the specified class through the reflection mechanism, and then instantiate the object

The above function of dynamically obtaining information and dynamically calling objects is called the reflection mechanism of the Java language

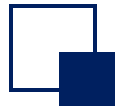
□ When to reflect

- Know the class to which any object belongs at runtime.
- Construct an object of any class at runtime.
- Know the member variables and methods that any class has at runtime.
- Call the methods and properties of any one object at runtime.



Class

- ❑ The Java runtime system always maintains a runtime type identifier for all objects
- ❑ It keeps track of the complete structure information of the class to which each object belongs, including package names, class names, implemented interfaces, methods and fields owned, etc., and the JVM uses this information to select the correct method to execute
- ❑ This information can be accessed using special Java classes
- ❑ Understanding of Class classes
 - The Class class can be understood as a type of class
 - A Class object is called a class type object



Class

```
import java.util.Date; // class first
public class Test {
    public static void main(String[] args) {
        Date date = new Date(); // object later
        System.out.println(date);
    }
}
```

2. The JVM looks up the Date.class file from the local disk and loads it into memory on the JVM

Date date = new Date

1. When we call new Date, the JVM loads Date.class

Date.class

JVM

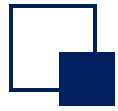
Date object space

3. Read the .class file into memory

Class object
(Stores information in the Date class)

3. Create a Class object

A class produces only one Class object



Class

□ Normal way

```
Date date = new Date();
```

The name class

new
instantiation

Instantiate
objects

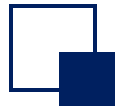
□ Reflection mode

Instantiate
objects

getClass()
method

Full package
class name

```
System.out.println(date.getClass()); // "class java.util.Date"
```

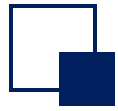


Obtaining Class Objects(1)

□ getClass() method

- The getClass() method in the Object class returns an instance of the Class type

```
public class getClassTest {  
    public static void main(String[] args) {  
        Student Harry = new Student("Harry Potter",11); // The Student class describes the student  
        System.out.println(Harry.getClass()); // The Class object describes the properties of a particular class  
        // output Class Student  
        System.out.println(Harry.getClass().getName()); // Class.getName() Returns the name of the class  
        // output Student  
    }  
}
```



Obtaining Class Objects(2)

□ Class.forName()

- Save the class name in a string

□ T.class

- T is an arbitrary Java type (may or may not be a class)

```
public class forNameTest {  
    public static void main(String[] args) throws ClassNotFoundException {  
        String className = "Reflect.Student";  
        Class cl1 = Class.forName(className); //Class.forName  
        Class cl2 = Student.class; //T.class  
    }  
}
```



Construct an instance of the class by reflection

❑ Method 1: Use Class.newInstance

- The newInstance method calls the default constructor (parameterless) to initialize the newly created object
- If the class does not have a default constructor, an exception is thrown

class has no parameterless constructor,
What if the class constructor is private?

```
Date date1 = new Date();  
Class dateClass2 = date1.getClass();  
Date date2 = dateClass2.newInstance();
```



Construct an instance of the class by reflection

❑ Method 2: Use Constructor's newInstance

- The construction method is obtained first through reflection and then called
- Get the constructor first, and then execute the constructor

❑ distinguish

- `Constructor.newInstance` can carry parameters
- `Class.newInstance` is parameterless



Construct an instance of the class by reflection

□ Get the constructor

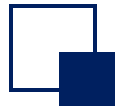
- Get all the "public" construction methods
 - `public Constructor[] getConstructors() { }`
- Get all constructs (including private, protected, default, public)
 - `public Constructor[] getDeclaredConstructors() { }`
- Get a "public" construction method that specifies the parameter type
 - `public Constructor getConstructor(Class... parameterTypes) { }`
- Get a "construction method" for the specified parameter type, which can be private, protected, default, or public
 - `public Constructor getDeclaredConstructor(Class... parameterTypes) { }`



Construct an instance of the class by reflection

❑ Use Constructor's newInstance to construct an instance of the class

```
public class ConTest {  
    public static void main(String[] args) {  
        Student Harry = new Student("Harry Potter", 11);  
        Class StudentClass = Harry.getClass();  
        Constructor con = StudentClass.getConstructor(String.class, int.class);  
        Student Ron = (Student)con.newInstance("Ron Weasley", 11);  
        System.out.println(Ron);  
    }  
}
```



Member variables are acquired and modified by reflection

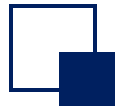
□ Get and modify member variables

- Get all publicly owned fields
 - `public Field[] getFields() { }`
- Get all fields (including private, protected, default)
 - `public Field[] getDeclaredFields() { }`
- Get a public field with a specified name
 - `public Field getField(String name) { }`
- Get a field with a specified name, which can be private, protected, default
 - `public Field getDeclaredField(String name) { }`
- Use the get method in the Field class to view the field values
- Use the set method in the Field class to modify the field value



Member variables are acquired and modified by reflection

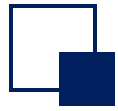
```
public class FieldTest {  
    public static void main(String[] args) {  
        Student Harry = new Student("Harry Potter", 11);  
        Class StudentClass = Harry.getClass();  
        Field f = StudentClass.getDeclaredField("name");  
        f.setAccessible(true);  
        Object v1 = f.get(Harry);  
        System.out.println(v1);  
        f.set(Harry, "The boy who lived");  
        System.out.println(Harry.getName());  
    }  
}  
// Calling f.set(Harry, newValue) can modify the attribute of the harry object to newValue
```



Acquisition of members by reflection

□ Acquisition of members by reflection

- Get all "public methods" (including the parent class and of course the Object class)
 - `public Method[] getMethods() { }`
- Get all member methods, including private (excluding inherited)
 - `public Method[] getDeclaredMethods() { }`
- Get a member method that specifies the method name and parameter type
 - `public Method getMethod(String name, Class<?>... parameterTypes)`



Call the member method of the reflection acquisition

Object invoke(Object obj, Object... args)

- The first argument is which object is going to call this method
- The first parameter is the argument passed when the method is called
- For static methods, the first parameter can be ignored and set to null

```
public class MethodTest {  
    public static void main(String[] args) {  
        Student Harry = new Student("Harry Potter", 11);  
        Method m = Student.class.getMethod("getName");  
        String n = (String) m.invoke(Harry);  
    }  
}
```



Reflection

□ advantage

- It is more flexible and can dynamically obtain instances of classes at runtime

□ disadvantage

- Performance bottlenecks: Reflection is equivalent to a series of interpretation operations that tell the JVM what to do, and the performance is much slower than direct Java code
- Security issues: The reflection mechanism breaks encapsulation, as the class's private methods and fields can be obtained and called through reflection.



Outline

- Why generics? What is generic?
- Generic classes, generic methods, and generic interfaces
- Generic wildcards
- Generic design - template method pattern
- Reflection
- Design security of singleton pattern

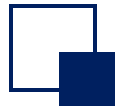


Singleton

❑ The purpose of a singleton is to ensure that there is only one instance of a class

- Instantiate an object when the class is loaded

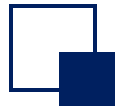
```
public class Singleton {  
    private static Singleton singleton = new Singleton();  
    private Singleton() {}  
  
    public static Singleton getInstance() {  
        return singleton;  
    }  
}
```

Breaks singleton using reflection

- ❑ Override access control in Java by `setAccessible(true)` in reflection, and then call a private constructor

```
public class SingletonTest {  
    public static void main(String[] args) {  
        Class objectClass = Singleton.class;  
        Constructor constructor = objectClass.getDeclaredConstructor();  
        constructor.setAccessible(true);  
        Singleton instance = Singleton.getInstance();  
        Singleton newInstance = (Singleton)constructor.newInstance();  
        System.out.println(instance == newInstance);  
    }  
}  
} // instance and newInstance are different instances of False
```



Resists reflection damage

```
public class Singleton {  
    private static Singleton singleton = new Singleton();  
    private Singleton(){  
        if(singleton != null) {  
            throw new RuntimeException("Singleton constructor prohibits calls by reflection");  
        }  
    }  
    public static Singleton getInstance(){  
        return singleton;  
    }  
}
```

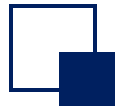


Problem

You are designing a small utility class called **TypePrinter**.

This utility should:

1. Accept **any type of object** (use generics).
2. Use **Java reflection** to print all **public method names** of that object's class.



Problem

You are designing a small utility class called **TypePrinter**.

This utility should:

1. Accept **any type of object** (use generics).
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```
class GenericBox<T> {  
    // TODO: declare field to store the value (type T)  
  
    // TODO: constructor to initialize the value  
  
    public T get() {  
        // TODO: return the stored value  
    }  
}  
  
class TypePrinter {  
    public static void printMethods(Object obj) {  
        // TODO: get the class of obj using reflection  
  
        // TODO: retrieve public methods  
  
        // TODO: loop and print all method names  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        GenericBox<String> box = new GenericBox<>("Hello Reflection!");  
  
        // Use reflection to print all methods of the stored object  
        TypePrinter.printMethods(box.get());  
    }  
}
```



Problem

```
import java.lang.reflect.Method;

class GenericBox<T> {
    private T value;    // field to store the generic value

    public GenericBox(T value) {    // constructor
        this.value = value;
    }

    public T get() {    // getter
        return value;
    }
}

class TypePrinter {
    public static void printMethods(Object obj) {
        Class<?> clazz = obj.getClass();    // get runtime class

        Method[] methods = clazz.getMethods();    // get all public methods

        System.out.println("Public methods of: " + clazz.getName());
        for (Method m : methods) {
            System.out.println(m.getName());    // print method name
        }
    }
}

public class Main {
    public static void main(String[] args) {
        GenericBox<String> box = new GenericBox<>("Hello Reflection!");

        // Use reflection to print all public methods of the stored object
        TypePrinter.printMethods(box.get());
    }
}
```



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